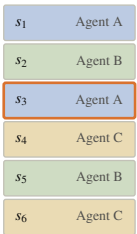


reference failures D_{fail}



unlabeled
failed trajectory τ



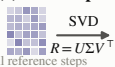
✗ task fails

no training · no step labels
no LLM prompting · one proxy



attention

(1) Fit the spectral band



R : all reference steps

SVD

$$R = U\Sigma V^T$$

once, on the unlabeled reference failures

keep a band C
 $= \{c_{\text{begin}}, \dots, c_{\text{end}} - 1\}$

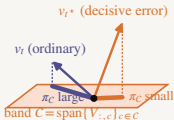
singular values of R , by index c



ordinary steps
dominate R , so
the band is their
common mode

(2) Spectral base score C

shadow = projection onto the band



$$\pi_C(v_t) = \frac{1}{|C|} \sum_{c \in C} \langle v_t, V_{:,c} \rangle^2$$

$$S(s_t) = 1 / (\pi_C(v_t) + \epsilon)$$

smaller projection $\Rightarrow$ larger score

rate every step by its projection onto the band

base score $S(s_t)$

$\arg \max S$ drifts past t^*
(downstream contamination)



$+\gamma B_i$

(3) Attention-guided rescoring

route evidence back to its source

earlier step $i \rightarrow$



dependency weights $w_{i,t}$
from attention; $\sum_{i < t} w_{i,t} = 1$

$$B_i = \frac{\sum_{t > i} w_{i,t} S(s_t)}{\sum_{t > i} w_{i,t}}$$

mean evidence of the
steps that depend on s_i

$$\tilde{S}(s_i) = S(s_i) + \gamma B_i$$

one pass over the original S ; a popular
step gains nothing unless its dependents fail

$$\hat{t} = \arg \max \tilde{S}(s_i) = s_3 = t^*$$

$$\hat{a} = a_{\hat{t}} = \text{Agent A}$$

rescored $\tilde{S}(s_i)$

